

Suppose f is strong convex with $mI \leq \nabla^2 f(x) \leq MI$

Show that the backtracking stopping condition holds for

$$0 < t < -\frac{\nabla f(x)^T \Delta x}{M \|\Delta x\|_2^2}$$

A:

$$(1) \nabla^2 f(x) \leq MI \Rightarrow f(x+t\Delta x) \leq f(x) + t\nabla f(x)^T \Delta x + \frac{M}{2} t^2 \Delta x^T \Delta x$$

$$(2) \text{ if } t \cdot (1-\alpha) \nabla f(x)^T \Delta x + \frac{M}{2} t^2 \Delta x^T \Delta x \leq 0 \quad \text{[stopping condition]}$$

$$(1) \text{ plus } (2) \Rightarrow f(x+t\Delta x) \leq f(x) + \alpha t \nabla f(x)^T \Delta x \quad \text{[stopping condition]}$$

$$\therefore t(1-\alpha) \nabla f(x)^T \Delta x + \frac{M}{2} t^2 \Delta x^T \Delta x \leq 0 \text{ is the stopping condition}$$

$$\Leftrightarrow 0 \leq t \leq -\frac{2(1-\alpha) \nabla f(x)^T \Delta x}{M \|\Delta x\|_2^2}, \quad \alpha \in (0, \frac{1}{2})$$

$$2(1-\alpha) \in (1, 2)$$

$$\Rightarrow \frac{-2(1-\alpha) \nabla f(x)^T \Delta x}{M \|\Delta x\|_2^2} \leq -\frac{\nabla f(x)^T \Delta x}{M \|\Delta x\|_2^2}$$

$$\Rightarrow 0 \leq t \leq -\frac{\nabla f(x)^T \Delta x}{M \|\Delta x\|_2^2} \quad \text{[stopping condition]}$$

B: let $t_0 = 1$, $t_{\text{stop}} = -\frac{\nabla f(x)^T \Delta x}{M \|\Delta x\|_2^2}$, $\beta < 1$, $\log \beta < 0$

$$\beta^k \cdot t_0 \leq t_{\text{stop}}$$

$$k \geq (\log t_{\text{stop}}) / \log \beta$$



Let $f(x) = \frac{1}{2}(x_1^2 + rx_2^2)$ with $r > 0$. Suppose that the gradient descent method is used with exact line search, starting at $x^{(0)} = (r, 1)$. Prove:

$$x_1^{(k)} = r \left(\frac{r-1}{r+1} \right)^k, \quad x_2^{(k)} = \left(-\frac{r-1}{r+1} \right)^k$$

How about Newton's descent?

Part A:

$$\textcircled{1} \nabla f(x) = [x_1, r x_2]^T$$

$$\textcircled{2} x^{(k+1)} = x^k + t \Delta x, \quad \Delta x = -\nabla f(x)$$

$$\Rightarrow x^{(k+1)} = x^k - t_k \nabla f(x^k)$$

$$\textcircled{3} \text{ let } g(t) = f[x^k - t \nabla f(x^k)] = \frac{1}{2} [(x_1^k - t x_1^k)^2 + r(x_2^k - t r x_2^k)^2]$$

$$\text{let } g'(t) = 0 \Leftrightarrow t_k = \frac{(x_1^k)^2 + r^2 (x_2^k)^2}{(x_1^k)^2 + r^2 (x_2^k)^2}$$

$$\textcircled{4} x^0 = (r, 1) \text{ substitute into } \textcircled{3}, \quad t_k = \frac{2}{1+r}$$

$$\textcircled{5} \begin{cases} x_1^1 = x_1^0 - t_0 \nabla f(x_1^0) = \left[\frac{2}{1+r}, 1 \right]^T r \frac{r-1}{r+1} \\ x_2^1 = x_2^0 - t_0 \nabla f(x_2^0) = \frac{1-r}{1+r} \end{cases}$$

$$\Rightarrow x_1^k = r \left(\frac{r-1}{r+1} \right)^k, \quad x_2^k = \left(-\frac{r-1}{r+1} \right)^k, \text{ Proved}$$

Part B. Newton's method:

$$x^{(k+1)} = x^k - [\nabla^2 f(x^k)]^{-1} \nabla f(x^k)$$

$$\nabla^2 f(x) = \begin{pmatrix} 1 & 0 \\ 0 & r \end{pmatrix}$$

$$\text{Newton Step: } x^{k+1} = x^k - \begin{pmatrix} 1 & 0 \\ 0 & 1/r \end{pmatrix} \begin{bmatrix} x_1^k \\ r x_2^k \end{bmatrix} = x^k - \begin{bmatrix} x_1^k \\ x_2^k \end{bmatrix} = 0$$

Newton's method converges in just one step to the optimal point $(0, 0)$



Let ΔX_{nsd} and ΔX_{sd} be the normalized and unnormalized steepest descent directions at x . Prove the following identities.

$$1. \nabla f(x)^T \Delta X_{nsd} = -\|\nabla f(x)\|_*$$

$$2. \nabla f(x)^T \Delta X_{sd} = -\|\nabla f(x)\|_*^2$$

Proof:

$$1. \Delta X_{nsd} = \arg \min_{\|\Delta x\| \leq 1} [\nabla f(x)^T \Delta x] \quad (1)$$

$$\|g\|_* = \sup_{\|x\| \leq 1} g^T x \quad (2)$$

$$\text{① according to (2), } \|\nabla f(x)\|_* = \sup_{\|\Delta x\| \leq 1} \nabla f(x)^T \Delta x$$

$$\inf_{\|\Delta x\| \leq 1} \nabla f(x)^T \Delta x = -\sup_{\|\Delta x\| \leq 1} (-\nabla f(x)^T \Delta x) = -\|\nabla f(x)\|_*, \text{ proved}$$

$$2. \Delta X_{sd} = \|\nabla f(x)\|_* \cdot \Delta X_{nsd} \quad (\text{by definition})$$

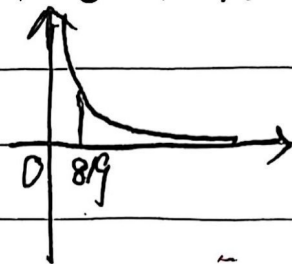
$$\nabla f(x)^T \Delta X_{sd} = \cancel{\|\nabla f(x)\|_*} \cdot \nabla f(x)^T \cdot \underbrace{\|\nabla f(x)\|_* \cdot \Delta X_{nsd}}_{\text{from (1)}} \therefore$$

$$\nabla f(x)^T \Delta X_{sd} = \nabla f(x)^T \cdot [-\|\nabla f(x)\|_*] = -\|\nabla f(x)\|_*^2$$



Show that $f(x) = \frac{1}{x^2}$ with domain $(0, 8/9)$ is self-concordant

$$\begin{cases} f'(x) = -\frac{1}{x^2} \\ f''(x) = \frac{2}{x^3} \\ f'''(x) = -\frac{6}{x^4} \end{cases}$$



definition of self-concordant:

$$|f'''(x)| \leq 2 f''(x)^{3/2} \text{ for all } x \text{ in domain}$$

$$\Leftrightarrow \left| \frac{-6}{x^4} \right| \leq 2 \left(\frac{2}{x^3} \right)^{3/2} \Leftrightarrow -6x^{-4} \leq 4\sqrt{2} x^{-\frac{3}{2}}$$

$$\Leftrightarrow \sqrt{x} \leq \sqrt{8/9}$$

$$\Leftrightarrow 0 \leq x \leq 8/9, \text{ Proved.}$$

